

Aren Desai

<https://arendesai.com> • github.com/ArenKDesai • arendesai@gmail.com

Education

Georgia Institute of Technology - <i>Online M.S. Computer Science (in progress)</i>	expected Summer 2029
University of Wisconsin - Madison - <i>B.S. Computer Science & Data Science</i>	Spring 2025
Google Cloud - <i>Professional Machine Learning Engineer Certification</i>	Spring 2025
FICO - <i>FICO DMP Certification</i>	Summer 2024

Technical Skills

Programming Languages: Python, SQL, R, Java, Rust

Modeling & Simulation: Monte Carlo simulation, gradient boosting, statistical modeling, OR-Tools, PyTorch, scikit-learn

Data Engineering: Polars, Pandas, DuckDB, Parquet, Apache Airflow, Docker

Cloud & Infrastructure: Microsoft Azure (Azure ML), Google Cloud Platform, Linux

Experience

Madison Gas & Electric - *Madison, WI* May 2024 – Present

Energy Supply Analyst

- Designed the daily price offer system for 3 battery energy storage resources (84 MW) in MISO's day-ahead market. Uses Monte Carlo simulation to identify optimal charge/discharge windows across multi-breakpoint strategies. Produces \$73,000+/yr in arbitrage revenue. Runs in production daily.
- Developed a probabilistic resource adequacy hour classifier covering a 120-hour horizon. Assigns risk probabilities to each hour based on predicted grid stress conditions. Informs operator pricing decisions and avoids \$150,150+/yr in capacity fees.
- Built an internal market simulation tool for traders, analysts, and generation planners. Enables rapid valuation and dispatch strategy comparison without full-market modeling runs.
- Built a multi-horizon day-ahead LMP forecaster for MISO pricing nodes. Uses Chronos2 foundation models with gradient boosting and 100+ features: weather, electrical load, fuel prices, and grid conditions. Reduced MAPE from 27% to 13% on a 5-day horizon.
- Lead the development of ML infrastructure, including the hiring of an intern and budgeting of cloud compute, deployment, and data pipelines.

Energy Supply and Trading Intern

- Accelerated MGE's MISO market simulation engine by 97.3% using SciPy optimization routines and OR-Tools, reducing runtime from hours to minutes.
- Bootstrapped MGE's ML infrastructure from zero. Drove the Azure ML platform buildout across DevOps and SecOps teams, securing compute resources and architecting the deployment environment.
- Prototyped the day-ahead LMP multi-horizon forecaster in PyTorch, which became the production system deployed as a full-time analyst.

Compeer Financial - *Sun Prairie, WI* Jan 2023 – Jan 2024

Data Analytics Intern

- Built a logistic regression classifier in scikit-learn to predict loan default risk across a 20,000-borrower dataset. Delivered findings and recommendations to the risk analysis team.
- Introduced PyTorch and modern ML tooling to Compeer's analytics stack through internal presentations and demos.

Leadership Experience

Google Developer Student Club Madison, WI
Finance Lead / Treasurer Sep 2023 – May 2024

- Managed club finances with UW Credit Union; organized tech talks and workshops
- Judged the 2024 CheeseHacks hackathon